

Tool Demo: Guided Label Propagation for Semi-supervised Detection of Affinities in Large-scale Information Sharing Networks

Introduction

This tool demo proposes a network analytical method, which has been named **guided label propagation** and its implementation as an available Python package. The tool can be considered a method for semi-supervised community detection in large-scale networks. It has been developed in the context of information sharing where actors on digital platforms are considered connected if they have shared the same messages (e.g. retweeting, re-sharing) or the same entities such as URLs, images, memes (Gruzdt et al., 2022) or similar distinguishable content features (Bruns et al., 2024).

The motivation behind the development of the tool is based on the current wide-spread use of network analysis and community detection. Network analysis is being used to find influential information spreaders (Bruns et al., 2021), their location in the information sharing landscape (Angus et al., 2023) and how well they cluster together with similar actors (Horne et al., 2019). In this endeavor community detection algorithms are often employed to map complex landscapes by identifying clusters of accounts with similar sharing behavior such as those often retweeting each other (Zhang et al., 2023), liking the same posts (Schmidt et al., 2017) or sharing the same links (Eady et al., 2020). Sometimes clusters are easily identified, such as in Glazunova et al., (2023), which maps the sharing of URLs to various Russian state-backed news sites on Facebook. However, at other times political information sharing patterns are not always simple to disentangle using standard community detection algorithms (e.g. Giglietto et al., 2022).

For complex networks containing tens of thousands of unknown accounts it can become difficult to ascertain what role they play in the network. **Guided label propagation** is a method to search for patterns pertaining specifically to categories of interest for the given study. The method has been used in practice to uncover actors engaging in the spread of misinformation and foreign interference propaganda [AUTHORS, 2024] in very large information sharing environment and in studies of political polarization [AUTHORS, 2023]. Rather than labelling nodes based on their belonging to clusters of a network partition, where the separation between clusters is determined by network structure (e.g. Blondel et al., 2008), **guided label propagation** instead relies on a small sample of *input nodes* with known affinities such as accounts belonging to political right- and left-wing actors. The labels of these known accounts (left and right wing) are then propagated throughout the network in order to uncover how strongly the rest of the accounts are affiliated with the position represented by those known accounts. The advantage of the approach lies in its ability to help researchers to identify the role of particular clusters in social media based information sharing networks by focusing on accounts' affinity towards known categories of interest rather than clusters that are initially arbitrary.

Previous work on semi-supervised community detection

This tool contributes by further developing a semi-supervised community detection technique to be used specifically for analyzing information sharing on social media. Compared to the vast amount of research on community detection methods in general there is a dearth of exploration of semi-supervised techniques. Most studies only use semi-supervised techniques to optimize the same labeling strategies used in most community detection methods (e.g. Liu et al., 2014). In contrast, Truong et al., (2024) proposes an approach where web domains are pre-labeled as credible or not credible sources and then used as input nodes for a bipartite network consisting of Twitter users and domains, they have linked to whereafter the labels from the web domains are propagated to the users in order to reveal those with a high affinity for sharing low credibility sources. A similar approach has also been explored by Rao et al., (2021) for political attitudes. Both works demonstrate the usefulness of the approach but without testing its wider applicability.

The general method proposed in this paper can theoretically be used for all types of information sharing behavior (retweeting, co-sharing hashtags, liking same content), however it is tested using networks based on social media accounts and their mutual sharing of URLs. The idea is for guided label propagation to be useful for finding patterns in networks containing sharing of very diverse types of content, which is the case with URLs that can link to news, entertainment, videos, commercial appearances, personal blogs etc. (González-Bailón et al., 2022; Bruns, 2023).

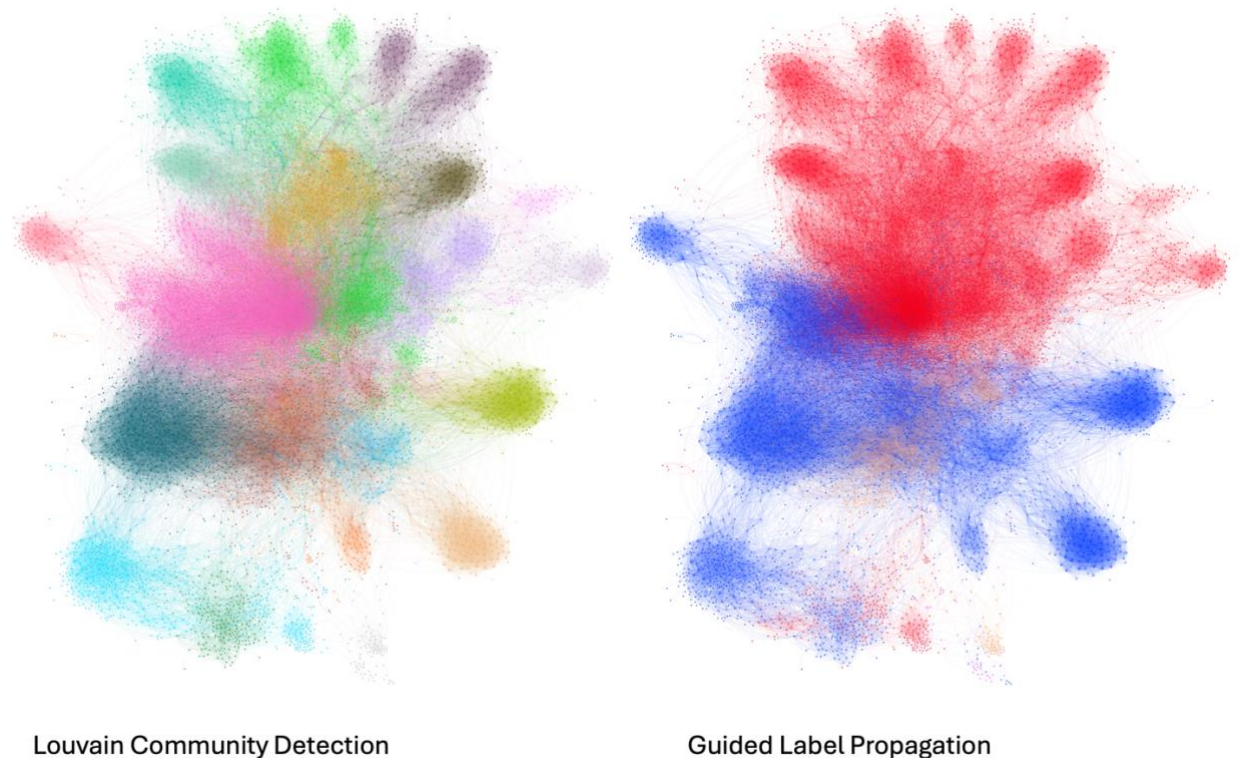
Exemplifying the tool: Detecting political affinities in networks based on mutual URL-sharing

The network constructed in this example builds on the assumption that if an account shares the same URL as another with a known political affiliation, then they are more likely to have the same political standpoint. Although, simply sharing a single URL to a right-wing outlet does not make an account a right-wing actor. The assumption is that over time, using enough data, accounts that slant more towards the right wing will share more of the same sources compared to left wing or other political entities, which is suggested by empirical research (e.g. Green et al., 2023).

The following example contains a network which is sampled from X (Twitter) and Facebook accounts that have at least once shared a URL from an account belonging to an alternative news outlet in Denmark (list found in Mayerhöffer, (2021)). However, the network also includes all non-alternative news URLs shared by the accounts as well as accounts that have then shared those non-alternative news URLs. Essentially it represents a network of 2 degrees of separation from Danish alternative news URLs. We want to use the known political agenda of the alternative news outlets to explore political sharing patterns. Because some accounts have shared only a few URLs linking to alternative news outlets and hundreds of links to other types of content we can utilize **guided label propagation** to infer political affinities by relying on multiple degrees of associations. For example, starting from those accounts that often link to

left-wing alternative news outlets we assume that a good portion of additional URLs they share are also motivated by their left-wing stance. By iteratively exploring such associations left-wing, right-wing and non-partisan alternative news labels can be propagated throughout the entire network.

Figure 1: Node labelling based on Louvain and Guided Label Propagation

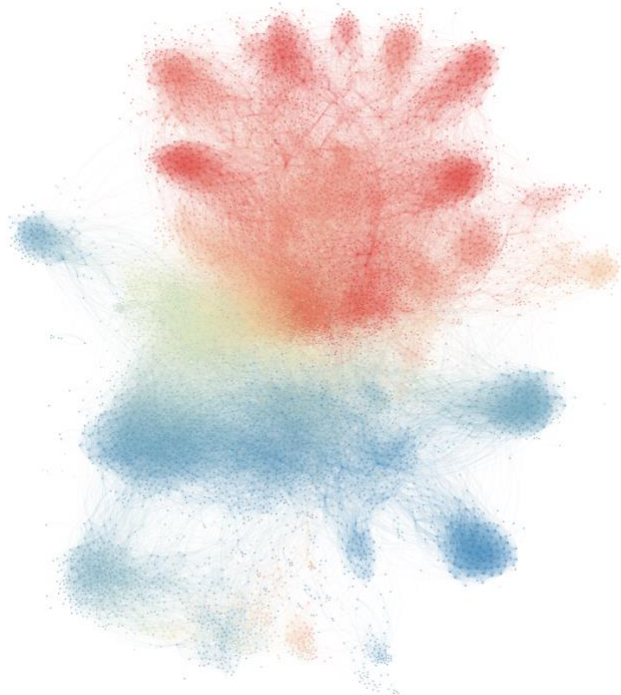


In the guided label propagation illustration nodes are colored accordingly: red = left-wing, blue = right-wing, yellow = non-partisan, purple = noise.

Figure 1 shows the labeling of nodes comparing a standard community detection technique (Louvain) with guided label propagation based on the three input labels right-wing, left-wing and non-partisan which were given to alternative news accounts with known political affiliation. The Louvain method detects a total of 21 clusters, which attests to the fact that there are factors other than the binary left-wing vs. right-wing classifications shaping URL sharing behavior. However, guided label propagation, in this case, helps to quickly give us an overview of the likely political orientation of accounts in the information sharing landscape which can help guide further investigation. Furthermore, guided label propagation also exports the normalized affinity scores based on the input labels, which can provide an even more precise overview of political orientation. Figure 2 shows node colors based on left-wing affinity (1 = very left-wing, 0 = not very left-wing), which can also help us identify nodes that have a balanced affinity towards both left and right-wing. For robustness guided label propagation also propagates a selection of randomly selected nodes to avoid overfitting for accounts that are very far away from any of the input nodes. Figure 3

shows “noisy areas” (purple) meaning the algorithm is less certain about the political affinity of those accounts.

Figure 2: Guided Label Propagation normalized left-wing scores



In the illustration red nodes are those with high left-wing scores, blue nodes with very low scores and light greenish ones those with balanced scores.

Figure 3: Guided Label Propagation normalized noise scores



The network contains 13.995 accounts total and uses only 27 accounts (those belonging to alternative news media with known political affiliation) to propagate left and right-wing labels to the rest. The results of the guided label propagation for the example are validated using a sample of 100 accounts belonging to politicians whose political affiliation is known. Despite most politicians not linking to alternative news sites, the **guided label propagation** is able to correctly label them by taking multiple degrees of association into account. In conclusion, this tool represents a viable approach for network analysis of complex information sharing behavior in cases where a more focused way of determining the role of clusters, compared to the completely unsupervised approach of traditional community detection algorithms, is needed.

Demonstrating the tool

Presenting the tool will include a live demonstration of importing the package into a jupyter notebook and running analyses. Starting from a raw csv edgelist based on the same data used in the example described in the previous section, the demonstration will show how the data is converted into a bi-partite network and then projected as a uni-partite network containing millions of edges between the accounts, carrying out backboning and finally the label propagation. All in less than a minute to exhibit the computational efficiency of the tool for working with very large-scale data. Finally, we take a quick look at the results of the **guided label propagation**, which can be exported both as a graph and as a table containing the nodes and their affinity scores pertaining to each category represented by the input nodes.

Specifications for the tool

- **Computationally efficient:** Implemented in Python using Polars and Networkit, which ensures multi-thread and multi-core calculations in all steps, which makes the tool very fast and scalable for large networks. Has been tested with networks of sizes ~4 billion edges (though an excess of 16 GB of RAM is required for networks of this size)
- **Multinomial and low sample size for input nodes:** The tool can propagate labels based a very small number of input nodes representing an arbitrary number of categories
- **Multiple types of graph data:** Works with bi-partite and uni-partite projections and both weighted and unweighted graphs as well as directed and undirected. In case of directed graphs, both labels pertaining to in-degree and out-degree are outputted. Although, multi-edge and dynamic graphs are NOT supported yet.
- **Backboning:** Implements multi-thread backboning to reduce noise in complex networks (Coscia, 2021).
- **Retains affinity scores for labelled nodes:** The output does not consist of only a single label for a single node. Affinity scores, as well as noise scores, pertaining to all nodes are outputted making it possible for researchers to

explore nodes with balanced affinity or top nodes that have strong local affinity towards a category that is globally insignificant.

- **Regularization and noise sensitivity hyperparameters:** Guided label propagation includes two hyperparameters, the first determines the weight placed on the distance to the input nodes. A sufficient weight (default=1) helps to ensure that the label propagation does not run amok and overtakes part of the network far away from the input nodes. The second determines how many nodes, calculated as a proportion to the number of input nodes, are labelled as noise nodes to counter the propagation of labels from the input nodes. If one is unsure how reliable some of the categories for the input nodes are, then a higher proportion of noise nodes can be included to prevent overfitting.
- **Link to Github:** <https://github.com/alterPublics/guidedLP> (A stable version is currently under development, so the tool might not work at the time of review)

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